

PROBABILITY AND STATISTICS

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BASIC CONCEPTS REQUIRED FOR UNDERSTANDING PROBABILITY

Experiment

The experiment means a planned activity or any process, which generates raw data.

Example: The tossing of a coin is considered an experiment.

Trial

A single performance of an experiment is called a trial.

Outcome

The result obtained from an experiment or a trial is called an outcome.

Random Experiment

An experiment, which produces different results even though, it is repeated a large number of times under essentially similar conditions are called a random experiment.

Example:

The tossing of a fair coin, the throwing of a balance die, drawing a card from a well-shuffled deck of 52 cards, selecting a sample, etc.

Sample Space

The set or collection of all possible outcomes of an experiment is called the sample space. It is denoted by the letter 'S'.

Example:

When we toss a coin, the sample space is $S = \{H, T\}$ or

When we roll a die, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$

Sample Point

Each outcome in a sample space is called a sample point or an element.

Example:

When we toss a coin, the sample space is $S = \{H, T\}$ and each of the element H (Head) and T (Tail) is a sample point.

TYPES OF EVENTS

Events

The possible outcome of any experiment is called an event. Thus an event is a subset of the sample space S . events are denoted by capital letters $A, B, C, D...$ etc.

Example:

When a coin is tossed, the sample space is $S = \{H, T\}$ and the subset $A = \{H\}$ is the event that a head occurs.

When two coins are tossed, the sample space is $S = \{HH, HT, TH, TT\}$ and the subset $B = \{HH, HT, TH\}$ is the event that at least one head appears.

Simple or Elementary Event

If an event consists of only one sample point, it is called simple or elementary event.

Example:

When two coins are tossed, the event $A = \{HH\}$ that head appears on both coins is simple event. When two dice are rolled, the event $B = \{(1, 1)\}$ that one appears on both the dice is simple event.

Compound Event

If the event consists of more than one sample points, it is called a compound event.

Example:

When two coins are tossed, the event $A = \{HH, HT, TH\}$ that at least one head appears is a compound event. When two dice are rolled, event $B = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$ that the sum of the two faces is 7, is a compound event.

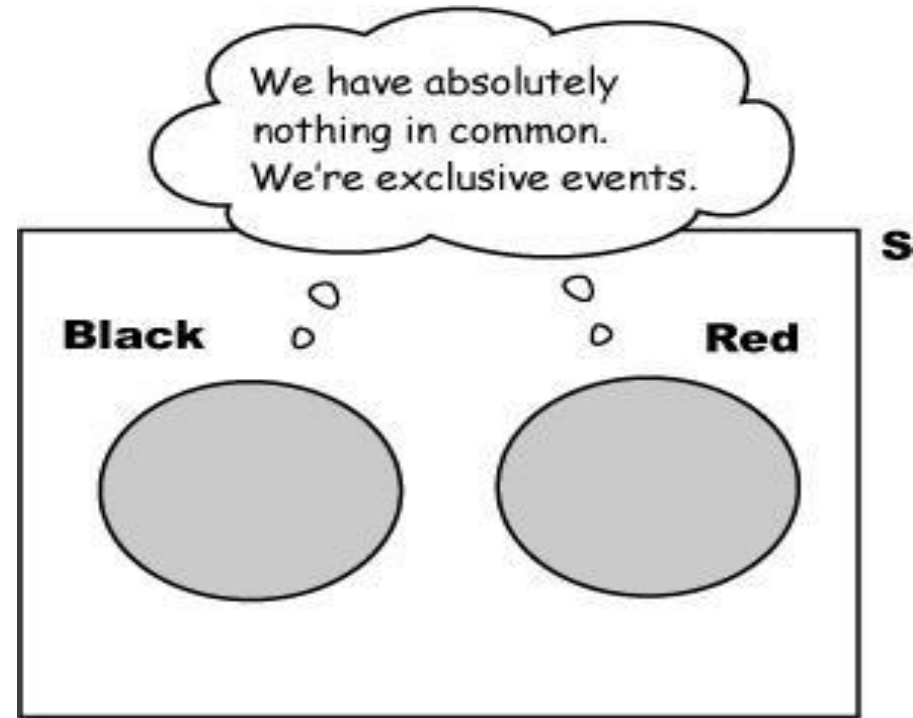
Mutually Exclusive Events

Two events are said to be mutually exclusive or **disjoint** if they both cannot occur at the same time.

(Or two events 'A' and 'B' are mutually exclusive if $(A \cap B) = \emptyset$; i.e. 'A' and 'B' have no elements in common) i.e. $P(A \cap B) = 0$

Example:

When we toss a coin, if the head occurs, the tail cannot occur. Thus, they are mutually exclusive. Similarly, if we roll a die and the face '4' occurs, the other faces cannot come up. Thus, all the six faces of a die are mutually exclusive.



Equally Likely Events

Two events are said to be equally likely, when one event is as likely to occur as the other.

Example:

If we toss a fair coin, the head is as likely to occur as the tail.

Exhaustive Events

Events are said to be exhaustive, when the union of mutually exclusive events is the entire sample space S .

Example:

If we toss a coin, the possible outcomes are a head and tail, there is no other possibility, namely, the coin will stand on edge etc.

Sure Event or Certain Event

An event that is certain to occur is called a certain or sure event. The sample space is a sure event.

Impossible Event

An event, which can never occur, is called an impossible event. The null set $\emptyset$ is an impossible event.

Independent Event

Two events are 'A' and 'B' are said to be independent events if occurrence of one does not affect occurrence or non-occurrence of the other. i.e.

$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(A / B) = P(A)$$

$$P(B / A) = P(B)$$

Example:

If we throw a pair of dice the occurrence of even number on one dice does not affect the occurrence or non-occurrence of an even number on the other dice.

Dependent Events

Two events are 'A' and 'B' are said to be dependent events if occurrence of one alters the chance of occurrence of the other event. i.e.

$$P(A \cap B) \neq P(A) \cdot P(B)$$

$$P(A / B) \neq P(A)$$

$$P(B / A) \neq P(B)$$

Example:

If we toss a coin, if head comes, tail cannot occur.

WHAT IS PROBABILITY

The word probability has two basic meanings

A numerical measure of uncertainty

A measure of degree of belief in a particular statement or problem.

DEFINITIONS OF PROBABILITY

Probability can be discussed from two points of view, the objective or subjective. The objective probability can be classified into the following categories, each of which is described as follows:

- a) The Classical or A Priori Definition of Probability
- b) The Relative Frequency or A Posteriori Definition of Probability
- c) The Axiomatic Definition of Probability
- d) Subjective or Personalistic Probability

THE CLASSICAL OR A PRIORI DEFINITION OF PROBABILITY

The classical or a priori definition of probability is given as follow:

If a random experiment can produce 'n' mutually exclusive or equally likely outcomes and if 'm' out of these outcomes are considered favorable to the occurrence of a certain event 'A', denoted by P (A).

Symbolically,

$$P(A) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$$

$$P(A) = \frac{m}{n}$$

THE RELATIVE FREQUENCY OR A POSTERIORI DEFINITION OF PROBABILITY

The relative frequency or a posteriori definition of probability is given as follows:

If a random experiment is repeated a large number of times, say 'n', under identical conditions and if an event A is observed to occur 'm' times, then the probability of the event A is defined as the limit of the relative frequency 'm/n' of 'n' tends to infinity.

Symbolically,

$$\lim_{n \rightarrow \infty} \frac{m}{n}$$

THE AXIOMATIC DEFINITION OF PROBABILITY

The axiomatic definition of probability is based on certain axioms, which must be satisfied by the probability of an event. Suppose a sample space 'S' has the sample points $E_1, E_2, \dots, E_i, \dots, E_n$. Each point is assigned a real number like

$P(E_1), P(E_2), \dots, P(E_i), \dots, P(E_n)$ which is called the weights or probabilities of the points. Probability of any point denoted by $P(E_i)$ must obey the following axioms:

$$0 \leq P(E_i) \leq 1$$

The probability of the event must lie between 0 and 1.

$$P(S) = 1$$

for the events which is as large as the sample space.

If A and B are mutually exclusive events, then

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$$

SUBJECTIVE OR PERSONALISTIC PROBABILITY

The subjective or personalistic probability is a measure of the strength of a person's belief regarding the occurrence of an event 'A'. The numerical measure of this confidence is called the subjective probability of occurrence of 'A'. This probability is based on the experience, intelligence and knowledge of the person who is determining the probability in some situation.

COUNTING SAMPLE POINTS

When the number of sample points in a sample space S is very large, it becomes very difficult to list them all and to count the number of points in the sample space S and in the subsets of S . We then need some methods or rules which help us to count the number of all sample points without actually listing them. A few basic rules frequently used in counting, are described below:

1. Rule of Multiplication
2. Rule of Permutation
3. Rule of Combination

RULE OF MULTIPLICATION

If a compound experiment consists of two experiments such that the first experiment has exactly m distinct outcomes and, if corresponding to each outcome of the first experiment there can be n distinct outcomes of the second experiment, then the compound experiment has exactly mn outcomes.

Example:

The compound experiment of tossing a coin and throwing a die together consists of two experiments: the coin tossing with two distinct outcomes (H, T), and the die-throwing with six distinct outcomes (1, 2, 3, 4, 5, 6). The total number of possible distinct outcomes of the compound experiment is therefore $2 \times 6 = 12$, as each of the two outcomes of the coin-tossing experiment can occur with each of the six outcomes of die-throwing experiment.

RULE OF PERMUTATION

A *permutation* is any ordered subset from a set of n distinct objects. The number of permutations of r objects, selected in a finite order from n distinct objects is denoted by the symbol ${}^n P_r$

The formulae is

$${}^n P_r = \frac{n!}{(n-r)!}$$

Example:

A club consists of four members. How many sample points are in the sample space when three officers: president, secretary and treasurer are to be chosen?

Solution:

$${}^n P_r = \frac{n!}{(n-r)!}$$

$${}^4 P_3 = \frac{4!}{(4-3)!} = \frac{4 \times 3 \times 2 \times 1}{1!} = \frac{4 \times 3 \times 2 \times 1}{1} = 24 \text{ (there are 24 possible sample points in the sample space)}$$

RULE OF COMBINATION

A *combination* is any subset of r objects, selected without regard to their order from a set of n distinct objects. The total number of such combinations is denoted by the symbol nC_r or $\binom{n}{r}$, (read “ n above r ”), where $r \leq n$.

The formulae is

$${}^nC_r = \frac{n!}{r! (n-r)!}$$

Example:

A three person committee is to be selected from a list of four persons. How many sample points are associated with the experiment?

Solution:

$${}^nC_r = \frac{n!}{r! (n-r)!}$$

$${}^4C_3 = \frac{4!}{3!(4-3)!} = \frac{4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = 4 \text{ (there are 4 possible sample points in the sample space)}$$

LAWS OF PROBABILITY

Law of Complementation

If ' $\bar{A}$ ' is the complement of an event 'A' relative to the sample space 'S', then

$$P(\bar{A}) = 1 - P(A)$$

ADDITION LAW OF PROBABILITY

Addition Law of Probability

(For Mutually Exclusive Events)

If 'A' and 'B' are two mutually exclusive events, then

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$$

ADDITION LAW OF PROBABILITY

Addition Law of Probability

(For not Mutually Exclusive Events)

If 'A' and 'B' are any two events, then

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

CONDITIONAL PROBABILITY

If 'A' and 'B' are two events, the probability that 'B' occurs given that 'A' has already occurred is called the conditional probability of 'B' given that 'A' has occurred and is denoted by $P(B/A)$. Symbolically,

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Example

We draw two cards without replacement from a well-shuffled pack of 52 cards and we want to find the probability that both are aces. The probability, that the second card is an ace given that the first card is an ace.

MULTIPLICATION THEOREM OF PROBABILITY (FOR DEPENDENT EVENTS)

If 'A' and 'B' are two events, then

$$P(A \cap B) = P(A) \cdot P(B/A)$$

or

$$P(A \cap B) = P(B) \cdot P(A/B)$$

MULTIPLICATION THEOREM OF PROBABILITY (FOR INDEPENDENT EVENTS)

If 'A' and 'B' are two independent events, then

$$P(A \cap B) = P(A) \cdot P(B)$$

BASIC SAMPLE SPACES REQUIRED FOR UNDERSTANDING THE CONCEPT OF PROBABILITY

Sample Space for one Coin

$$S = \{ H, T \}$$

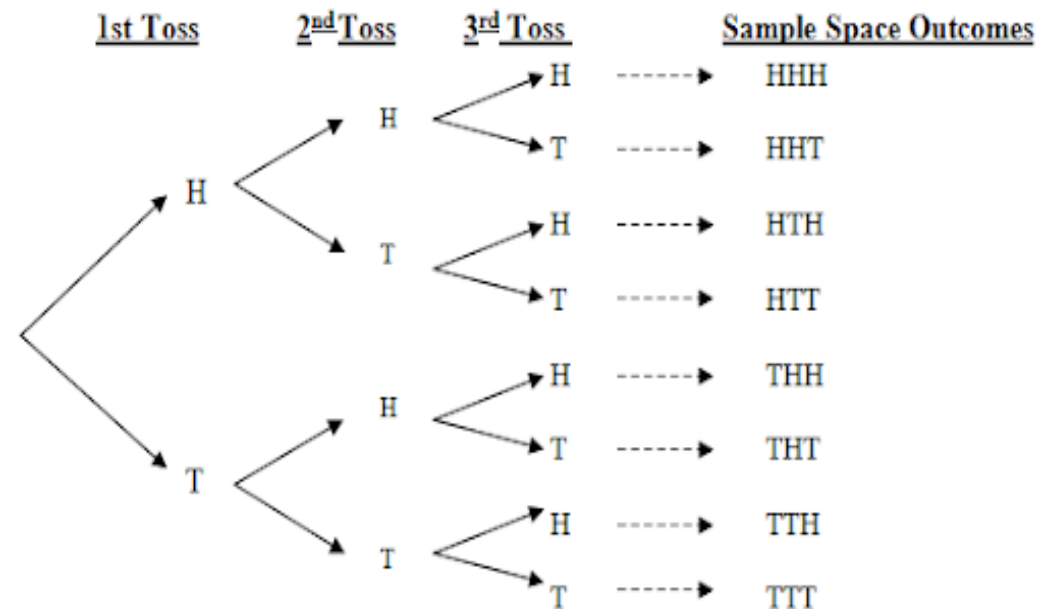
Sample Space for two Coins

Table for Tossing 2 Coins

	Heads	Tails
Heads	H, H	H, T
Tails	T, H	T, T

Sample Space: $\{HH, HT, TH, TT\}$

Sample Space for three Coins

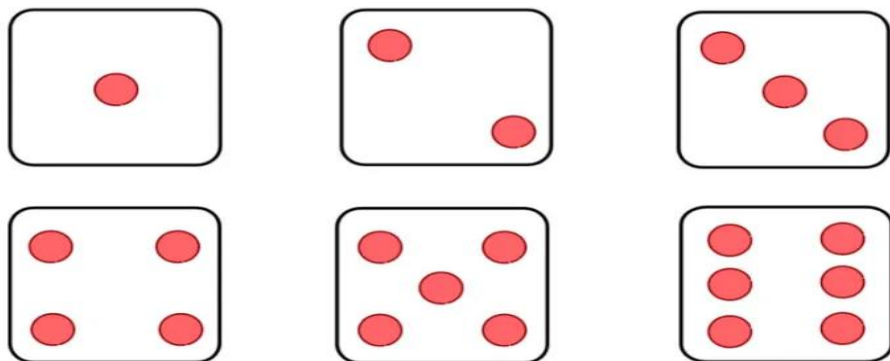


SAMPLE SPACE FOR SINGLE AND TWO DICE

Sample space for single die

$$S = \{ 1, 2, 3, 4, 5, 6 \}$$

Sample Space for Rolling a Die:



6 outcomes

Sample space for two dice

		Second throw					
		1	2	3	4	5	6
First throw	1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
	2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
	3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
	4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
	5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
	6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

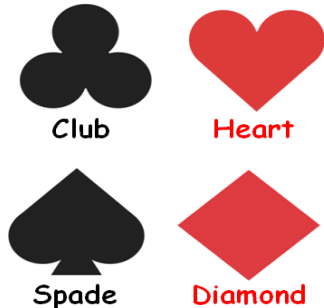
PLAYING CARDS DISTRIBUTION

Basic Stuff

Before we enter in to the stuff probability, we must be aware of the pack of 52 playing cards.

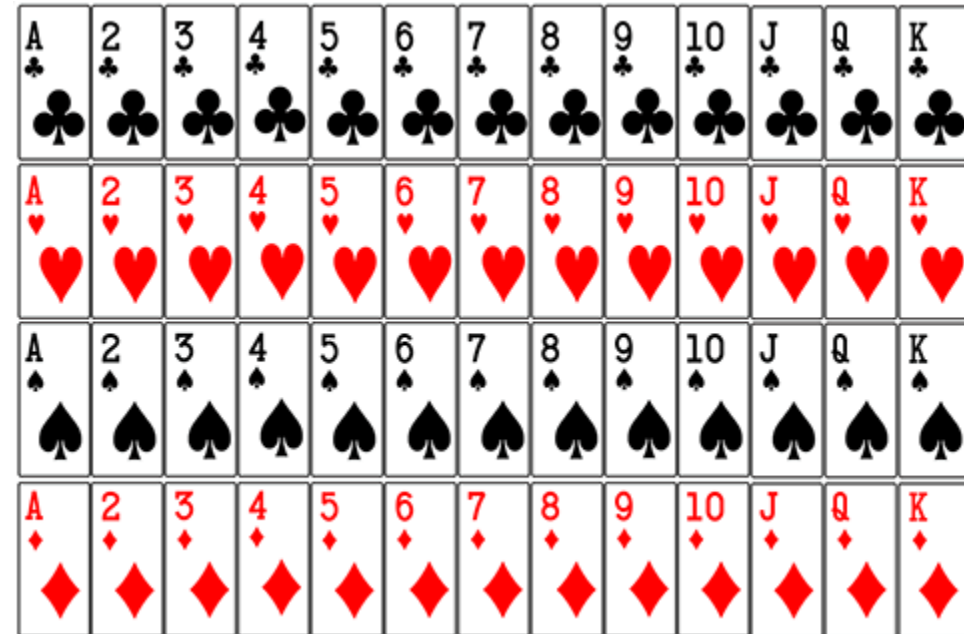
There are four types of cards in the pack of 52 cards. Two types in black color and other two types in red color.

The four different types of cards are shown in the picture given below.



In the above four types, we will have 13 cards in each type.

It is shown in the picture given below.



PRACTICE PROBLEMS

Question:

Find the probability of getting a tail in a single flip of a coin.

Solution:

The sample space S is given by.

$$S = \{H, T\}$$

Let A be the event "of getting a tail".

$$A = \{T\}$$

We use the formula of the classical probability.

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{2}$$

Question:

Two coins are tossed, find the probability that two heads are obtained.

Solution:

The sample space S is given by.

$$S = \{HH, HT, TH, TT\}$$

Let A be the event "two heads are obtained".

$$A = \{HH\}$$

We use the formula of the classical probability.

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{4}$$

Question: In case of rolling a die, what is the probability of getting:

a) an even number

b) not even number

c) Number greater than 4

Solution:

The required sample space is

$$S = \{ 1, 2, 3, 4, 5, 6 \}$$

a) probability of an even number

Let A be the event that represents the probability of an even number

$$A = \{ 2, 4, 6 \}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

b) Probability not even number

Let B be the event that represents the probability of not an even number

By using law of complementation

$$P(B) = 1 - P(A)$$

$$P(B) = 1 - \frac{1}{2} = \frac{1}{2}$$

Question:

Two dice are rolled, find the probability that the sum is

- a) equal to 1
- b) equal to 4
- c) less than 13

Solution:

The sample space S of two dice is shown below.

$S = \{ (1,1),(1,2),(1,3),(1,4),(1,5),(1,6)$
 $(2,1),(2,2),(2,3),(2,4),(2,5),(2,6)$
 $(3,1),(3,2),(3,3),(3,4),(3,5),(3,6)$
 $(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)$
 $(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)$
 $(6,1),(6,2),(6,3),(6,4),(6,5),(6,6) \}$

Part a) Let A be the event "sum equal to

1". There are no outcomes which correspond to a sum equal to 1, hence

$$P(A) = \frac{n(A)}{n(S)} = \frac{0}{36} = 0$$

Part b) Let B be the event "sum equal to 4". Three possible outcomes give a sum equal to 4

$B = \{(1,3),(2,2),(3,1)\}$, hence.

$$P(B) = \frac{n(B)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

c)) Let C be the event "sum is less than 13". All possible outcomes for event C is entire sample space because the maximum sum is 12, hence.

$$P(C) = \frac{n(C)}{n(S)} = \frac{36}{36} = 1$$

Question:

A card is drawn at random from a well shuffled pack of 52 cards. What is the probability that the drawn card is king ?

Solution :

Let A be the event of drawing a king card.

There are 4 king cards in total.

Then,

$$n(A) = 4$$

And also

$$n(S) = 52$$

Required probability is

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{4}{52}$$

$$P(A) = \frac{1}{13}$$

So, the probability of getting a kind card is $\frac{1}{13}$.

Question:

A card is drawn at random from a well shuffled pack of 52 cards. What is the probability that the card is neither a spade nor a Jack ?

Solution :

Let A be the event of drawing a card that is neither spade nor Jack.

There are 13 spade cards in a pack of 52 cards.

(Jack in spade is already included in these 13 cards)

In the remaining 3 types (club, heart, diamond), there are 3 jack cards.

Number of spade cards and jack cards is

$$= 13 + 3$$

$$= 16$$

So, no. of cards which are neither spade nor Jack are

$$= 52 - 16$$

$$= 36$$

Then,

$$n(A) = 36$$

And also

$$n(S) = 52$$

Required probability is

$$P(A) = \frac{n(A)}{n(S)}$$

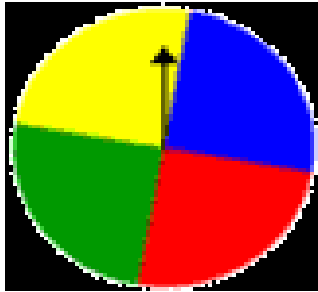
$$P(A) = \frac{36}{52}$$

$$P(A) = \frac{9}{13}$$

So, the required probability is $\frac{9}{13}$.

Question:

A spinner has 4 equal sectors colored yellow, blue, green, and red. What is the probability of landing on red or blue after spinning this spinner?

**Solution:**

$$P(\text{Red}) = \frac{1}{4}$$

$$P(\text{Blue}) = \frac{1}{4}$$

$$P(\text{Red or Blue}) = P(\text{Red}) + P(\text{Blue})$$

$$P(\text{Red or Blue}) = \frac{1}{4} + \frac{1}{4}$$

$$P(\text{Red or Blue}) = \frac{2}{4}$$

$$P(\text{Red or Blue}) = \frac{1}{2}$$

Example:

If you take out a single card from a regular pack of cards, what is probability that the card is either an ace or spade?

Solution:

Let A be the event of picking an ace and B be the event of picking a spade.

$$P(A) = \frac{4}{52}$$

$$P(B) = \frac{13}{52}$$

The two events are not mutually exclusive, as there is one favorable outcome in which the card can be both an ace and spade.

$$P(A \cap B) = \frac{1}{52}$$

Now, by using the addition law of probability for not mutually exclusive events which is as follows:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52}$$

$$P(A \cup B) = \frac{16}{52}$$

$$P(A \cup B) = \frac{4}{13}$$